Coding & Solution

Date	28 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Coding & Solution

The Smart Lender Loan Eligibility Prediction System is developed using Python, Flask, and Machine Learning techniques to predict whether a loan applicant is eligible for approval. The project includes data preprocessing, exploratory data analysis, model training, and a user-friendly web interface. The code is modular, well-structured, and includes error handling to ensure reliable performance. The solution successfully addresses the problem of automating loan approval prediction and can be further enhanced with additional features and cloud deployment.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/BrundhaMadiraju/Smart_Lender/tree/main
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy
Key Features Implemented	Loan eligibility prediction, data preprocessing, EDA analysis, trained ML model, user-friendly web interface, confidence score prediction
Pending / Incomplete Features	Database integration, user authentication, deployment on cloud platform
Setup / Run Instructions	Install dependencies using `pip install -r requirements.txt`, run `python app.py`, and open `http://127.0.0.1:5000/` in the browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The Smart Lender project follows good coding practices with a clean and modular structure. The application successfully predicts loan eligibility using machine learning and provides a simple, user-friendly web interface for users.